

Shea Loughnane

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Education

University of Michigan

Bachelor of Science in Computer Science

Ann Arbor, MI

May 2025

- **GPA:** 3.52
- **Course Highlights:** Data Structures & Algorithms, Software Engineering, Artificial Intelligence, Computer Security, Election Cybersecurity, Statistics & Data Analysis, Money & Banking
- **Honors:** Victor's Award (merit-based scholarship), University Honors, National Collegiate Rugby Scholastic All-American

Work Experience

Sentry Rugby Recruiting (sentryrugby.com)

Founder & Software Engineer

Boston, MA

July 2025 - Present

- Built a full-stack MERN web application to modernize college rugby recruiting, replacing manual workflows with semi-automated email outreach, stage-based recruit tracking, and a custom analytics dashboard
- Designed and implemented a scalable MongoDB schema using Mongoose, leveraging embedded documents for recruits, referenced relationships for teams and users, and schema-level validation to maintain data integrity
- Developed a modular REST API with Express.js, implementing JWT authentication, role-based access control, and middleware-based input validation to ensure secure, reliable data exchange
- Created a responsive React frontend with modular components and state-driven UI updates, integrating REST API data to deliver real-time insights and a seamless user experience across devices

Irrational Capital

Financial Research & Data Engineering Contractor

Remote

May 2023 - Present

- Built a Python tool to automate ETF portfolio construction, reducing manual effort by several hours per quarter and improving transparency and traceability of security selections and weightings
- Maintained and enhanced data pipelines leveraging Pandas, SQL, and BigQuery for data cleaning, validation, and integration, improving data quality and reliability for downstream analysis

Baird Augustine

Investment Banking / Hedge Fund Analyst

Remote

June 2024 - June 2025

- Supported multiple client engagements by preparing detailed company valuations, pitch materials, and due diligence reports, enabling senior management to make strategic, data-driven decisions
- Identified, evaluated and presented investment opportunities through comprehensive quantitative and qualitative research, driving maximized risk-adjusted returns for Limited Partners

QuickTap Rugby Analytics

Software Engineer

Remote

June 2024 - July 2024

- Spearheaded the development of interactive game data visualizations using Plotly.js, including a horizontal 100% stacked bar chart for head-to-head stats and a goal kicking heat map, empowering teams with deeper performance insights

Technical Skills

- **Languages:** Python, C++, C, Java, JavaScript, HTML, CSS, SQL
- **Frameworks & Libraries:** React, Node.js, Express.js, Mongoose, Axios, Jest
- **Tools:** Git, AWS, Cloudflare, BigQuery, MongoDB, Vite, Wireshark, Figma

Additional

- **University of Michigan Men's Rugby:** Captain, 1st Team All-Big Ten, Midwest All Star, Free Jacks U23
 - Awarded the opportunity to live and play rugby in Wellington, New Zealand for three months in summer 2023

Shea Loughnane
74 Sterling Road
Westwood, MA 02090
October 24, 2025

Michigan Robotics
2505 Hayward Street,
Ann Arbor, MI 48109

Dear Hiring Manager:

I am excited to apply for the Software Engineer position at Michigan Robotics. I graduated from the University of Michigan with a B.S. in Computer Science and am passionate about building software that solves real-world problems. Michigan Robotics' mission to create smart machines that serve society resonates deeply with me, and I am eager to contribute to your team by developing products that are both technically robust and impactful for members of the Robotics department.

As the founder and sole engineer of Sentry Rugby Recruiting, I designed and built a full-stack MERN application that modernizes collegiate rugby recruiting. I implemented a scalable MongoDB schema, modular REST API, and responsive React frontend to deliver a platform that automates workflows and provides actionable insights for teams. This project challenged me to think critically about architecture, data integrity, and user experience, and strengthened my ability to build clean, efficient, and intuitive end-to-end systems.

At Irrational Capital, where I worked as a Financial Research & Data Science Contractor, I developed a Python automation tool that replaced the manual ETF portfolio construction process. By collaborating closely with investment analysts, I tailored the tool to their specific needs, saving several hours of work per quarter and improving data accuracy and traceability. This experience broadened my technical versatility and reinforced my commitment to thoughtful, user-centered engineering.

Beyond my technical work, serving as Captain of the University of Michigan Men's Rugby Team honed my leadership, collaboration, and adaptability. I thrive in dynamic, high-pressure environments, whether coordinating on the field or solving complex engineering challenges with a diverse team.

I would be thrilled to bring my full-stack expertise, problem-solving mindset, and collaborative spirit to Michigan Robotics. Thank you for considering my application. I welcome the opportunity to discuss how my background and passion align with your team's goals, and can be reached at (781) 249-2436 or shealoughnane@gmail.com.

Sincerely,
Shea Loughnane